Content Provenance for the AI Era: Cryptographic Approaches to Establishing Trust in Digital Text

Dr. Elena Vasquez, Prof. James Chen

As large language models proliferate, the ability to distinguish humanauthored content from AIgenerated text has become a critical challenge for publishers, regulators, and the public. Traditional AIdetection tools rely on statistical classifiers whose accuracy degrades with each new model generation. This paper presents an alternative paradigm: cryptographic content provenance. Rather than asking "was this written by AI?", provenance systems answer "who published this, and has it been altered?" We survey the C2PA standard for text, introduce lightweight manifest modes that reduce embedding overhead by 90%, and evaluate zerowidth Unicode steganography for persentence integrity verification. Our results demonstrate that invisible cryptographic signatures can be embedded in plain text with negligible size increase while surviving common transformations including copypaste, format conversion, and content management system processing. 1. Introduction

The rapid advancement of generative AI has created an unprecedented crisis of trust in digital content. Publishers who invest in original journalism find their work indistinguishable from AIgenerated imitations. Regulatory frameworks like the EU AI Act and California SB 942 now mandate transparency around AIgenerated content, yet enforcement remains elusive without reliable technical mechanisms. Content provenance offers a fundamentally different approach to the authenticity problem. Instead of attempting to classify content after the fact, provenance systems embed cryptographic proof at the point of creation. The Coalition for Content Provenance and Authenticity (C2PA) has established an open standard for this purpose, originally designed for images and video. Extending this framework to text presents unique challenges: text is routinely copied, reformatted, and excerpted in ways that would destroy traditional watermarks. In this paper, we present three key contributions. First, we describe how the C2PA manifest structure can be adapted for text documents using invisible Unicode encoding. Second, we introduce lightweight and minimal manifest modes that dramatically reduce the overhead of provenance embedding. Third, we evaluate zerowidth character steganography for persentence integrity verification, enabling finegrained attribution even when documents are partially quoted or excerpted. 2. Background and Related Work

The C2PA specification defines a manifest structure containing claims, assertions, and cryptographic signatures that establish the provenance of digital assets. Originally designed for binary media formats, the standard uses CBORencoded data structures signed with X.509 certificates. For images, these manifests are embedded in metadata containers such as EXIF or XMP. Text documents lack equivalent metadata containers, necessitating alternative embedding strategies. Prior work on text watermarking has explored statistical approaches, including synonym substitution and syntactic transformations. These methods alter the content itself, raising concerns about fidelity and detectability. Unicode steganography, by contrast, embeds information in invisible characters that do not change the rendered text. Previous Unicode steganography schemes used variation selectors or zerowidth characters but lacked cryptographic integrity guarantees and standardized verification workflows. Our approach builds on these foundations by combining C2PAcompliant manifest structures with Unicode steganography, creating a system where provenance metadata is both invisible and cryptographically

verifiable. This allows publishers to sign content at creation time while preserving the exact text that readers see. 3. Methodology

3.1 C2PA Manifests for Text

We encode C2PA manifests as CBOR binary data, then represent this binary as sequences of invisible Unicode characters. The full manifest includes the claim generator identity, content assertions, action history, and an Ed25519 digital signature. This data is appended to the document text as a contiguous block of tag characters from the Unicode Supplementary Special-purpose Plane, specifically the range U+E0001 to U+E007F.

The full manifest approach provides maximum provenance detail but introduces significant character overhead. For a typical 500-word article, the full C2PA manifest adds approximately 15,000 invisible characters, representing a 40-60% increase in document size. While these characters are invisible to readers, they increase storage requirements and may be stripped by certain text processing pipelines. 3.2 Lightweight and Minimal Manifests

To address the overhead of full manifests, we introduce two reduced-footprint alternatives. The lightweight manifest replaces the full CBOR payload with a 128-bit UUID pointer. The complete manifest is stored server-side, and the UUID serves as a lookup key for verification. This reduces the per-document embedding to approximately 1,500 invisible characters: a 90% reduction compared to the full manifest. The minimal manifest mode extends this concept to per-segment embedding. Each sentence receives its own UUID pointer, while a single document-level C2PA wrapper provides the overarching provenance context. This enables fine-grained attribution: if a single sentence is copied from a signed document, the UUID embedded within that sentence can be used to trace it back to its origin, even without the surrounding document context. 3.3 Zero-Width Steganographic Embedding

For environments requiring maximum compatibility, we developed a zero-width character encoding scheme using four carefully selected Unicode characters: Zero-Width Non-Joiner (U+200C), Zero-Width Joiner (U+200D), Combining Grapheme Joiner (U+034F), and Mongolian Vowel Separator (U+180E). These four characters form a base-4 encoding alphabet, allowing us to represent arbitrary binary data in sequences of invisible characters. Each sentence receives a 128-character signature consisting of a 16-byte UUID and a 16-byte HMAC-SHA256 truncated digest. The HMAC is computed over the sentence text using a per-organization signing key, binding the sentence content to its provenance record. At the document level, a full HMAC signature covers the concatenation of all sentence-level signatures, creating a hierarchical integrity structure analogous to a Merkle tree. This approach offers several advantages over variation selector encoding. The four selected characters render invisibly across all major platforms including Microsoft Word, Google Docs, web browsers, and PDF viewers. Our testing confirmed that variation selectors, while offering higher information density, render as visible boxes in Microsoft Word, making them unsuitable for enterprise document workflows. 4. Results

We evaluated our embedding approaches across four dimensions: size overhead, platform compatibility, verification accuracy, and robustness to common transformations. Testing was conducted on a corpus of 5,000 news articles ranging from 200 to 3,000 words. The full C2PA manifest mode achieved 100% verification accuracy with an average size increase of 47%. Lightweight UUID mode reduced this overhead to 4.2% while maintaining 100% verification accuracy for documents where the server-side manifest remained accessible. Minimal UUID per-sentence mode added 6.8% overhead but enabled individual sentence verification. Zero-width

steganographic embedding added 3.1% overhead per sentence with full platform compatibility. Robustness testing demonstrated that all embedding modes survived copy-paste operations across Chrome, Firefox, Safari, Microsoft Word, and Google Docs. PDF export from all tested applications preserved the invisible characters. Content management systems including WordPress and Drupal preserved embeddings when configured to allow Unicode pass-through. The primary failure mode was aggressive text sanitization that strips non-printable characters, which affected 12% of tested CMS configurations. 5. Conclusion

Cryptographic content provenance provides a robust, standards-based alternative to probabilistic AI detection. By embedding invisible signatures at creation time, publishers can establish verifiable proof of authorship that persists through the content distribution lifecycle. Our lightweight and zero-width embedding modes make this practical for real-world deployment, with overhead levels acceptable for production content pipelines. Future work will focus on three areas: extending the approach to structured document formats including HTML and PDF, developing browser extensions for automatic verification, and establishing interoperability with the broader C2PA ecosystem for images and video. As regulatory requirements for content transparency increase, cryptographic provenance will become an essential tool for maintaining trust in the digital information ecosystem. References

Coalition for Content Provenance and Authenticity. "C2PA Technical Specification v2.1." c2pa.org, 2024. Kirchner, J. et al. "A Watermark for Large Language Models." Proc. ICML 2023. European Parliament. "Regulation (EU) 2024/1689: Artificial Intelligence Act." Official Journal of the EU, 2024. Kaptchuk, G. et al. "Meteor: Cryptographically Secure Steganography for Realistic Distributions." ACM CCS 2021. Unicode Consortium. "Unicode Standard Annex #9: Unicode Bidirectional Algorithm." unicode.org, 2023.